Taipei Housing Prices

Explaining Taipei Housing Prices: Do Price Relationships Differ Across Location, Time, and Market Segments?

1. Research Question and Project Type

Project type: Primarily descriptive/associational, with a secondary predictive extension. We do not make causal claims.

Main question: Which property and location characteristics are most strongly associated with residential price per ping and total transaction price in Taipei, and are these relationships stable across districts, time periods, and market segments?

- Does building age have the same relationship with price across districts?
- Are elevator and parking premiums similar for lower- and higher-priced properties?
- Have district price differences changed between 2019 and 2026?

2. Hypothesis and Reasoning

Hypothesis: Newer properties, properties with elevators or parking, and properties in higher-priced districts will tend to have higher price per ping. We also expect these relationships to vary across districts and over time.

Reasoning: Taipei districts differ in accessibility, neighborhood characteristics, and housing stock. Building age, elevator access, parking, and property type may therefore be valued differently in different local markets and periods.

3. Data

Source: Taipei City Actual Price Weekly Data (臺北市實價周報), provided by the Taipei City Government Department of Land Administration. Taipei Open Data

Coverage: The official dataset is an open, raw CSV covering March 2019 to October 14, 2026 and is updated every seven days. One row represents a reported real-estate transaction record before our residential-sale filtering.

Planned sample: Residential sale transactions only. The exact final number of observations will be reported after filtering out rentals, pre-sale records where inappropriate, duplicates, and unusable/missing records.

Role	Variables	Purpose
Outcomes	Price per ping; total transaction price	Measure housing price
Location/time	District; transaction date/year	Compare places and trends
Property	Area; rooms; building type; floor; total floors	Physical characteristics
Building features	Completion date/age; elevator; parking; land-use category	Amenities and building condition

4. Planned Analysis and Evidence

- Clean the data: keep appropriate residential sales, convert dates/numeric fields, construct building age, standardize categories, and document missing values/outliers.
- Describe the market with price distributions, district/building-type comparisons, and 2019–2026 price trends.
- Examine associations between price and area, age, floor, rooms, elevator, parking, building type, and district.
- Estimate a multiple regression of price per ping on district, age, area, floor, building type, elevator, parking, rooms, and transaction year.

- Test a small number of meaningful heterogeneity patterns across district, time, and price segment, chosen after exploratory analysis.
- Secondary prediction: compare a simple model with richer models and evaluate prediction error on held-out transactions.

5. Brief Motivation from Prior Studies

Previous Taipei housing studies have examined specific location-related factors such as hospital proximity, MRT accessibility, flood risk, road conditions, and neighborhood characteristics. For example, Peng and Chiang (2015) find that hospital proximity has a non-linear relationship with property prices, describing the preferred location as "**close-but-not-too-close**." Liou et al. (2016) also find that housing-price growth differs by location and building type, while the effect of MRT distance itself is not consistently significant.

Our project does not aim to claim that these factors are completely new. Instead, our incremental contribution is to use recent **2019–2026 transaction data** to examine a broader set of property characteristics together and test whether their relationships with price differ across **districts, time periods, and market segments**.

Our **baseline analysis** will use variables already available in the housing transaction dataset, such as district, building age, area, floor, building type, elevator, parking, rooms, and transaction year.

Variables emphasized in previous research, such as **MRT accessibility, road/traffic conditions, hospitals, flood risk, or nearby amenities, will be treated as possible extensions rather than required parts of the project**. If reliable external data can be joined efficiently, we may add one or two of these factors to an extended model and test whether they improve explanation or prediction beyond the baseline housing characteristics.

This lets us compare:
Baseline model:
property characteristics + district + time
Possible extended model:
baseline variables + selected accessibility/neighborhood factors
Therefore, the project can still be completed using the official housing transaction data alone, while external location variables provide a possible additional analysis if time and data availability allow.

References

Peng, T.-C., & Chiang, Y.-H. (2015). The non-linearity of hospitals' proximity on property prices: experiences from Taipei, Taiwan. Journal of Property Research, 32(4), 341–361. https://doi.org/10.1080/09599916.2015.1089923

Liou, F.-M., Yang, S.-Y., Chen, B., & Hsieh, W.-P. (2016). The effects of mass rapid transit station on the house prices in Taipei: the hierarchical linear model of individual growth. Pacific Rim Property Research Journal, 22(1), 3–16. https://doi.org/10.1080/14445921.2016.1158938

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